Aenōn's Theory: Mathematical Model of Reality

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Abstract

This document presents Aenōn's Theory, a comprehensive mathematical framework proposing to describe physical reality across all fields through a dimensionless parameter system and unified structural equations. The theory suggests that reality can be understood as a recursive phase-encoded excitation of quantized fields operating within an entropy-bound spacetime manifold, governed by conserved symmetries and expressed through fundamental dimensionless constants.

Introduction

The quest to understand the fundamental nature of reality has driven scientific inquiry for millennia. Aenōn's Theory proposes that at its most compressed form, physical reality can be described by a dimensionless parameter framework that unifies quantum mechanics, general relativity, thermodynamics, and information theory into a single coherent model.

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1. Dimensionless Constants: The Foundation of Physical Laws

The theory posits that observable physical laws emerge from eight fundamental dimensionless constants, empirically validated across multiple experimental domains:

Name	Symbol	Value
Fine-structure constant	α	$7.2973525693 \times 10^{-3}$
Gravitational coupling	αG	5.9×10^{-39}
Strong coupling constant	αs	≈ 0.1181
Weak mixing angle	$\sin^2\theta W$	0.23126
Electron–proton mass ratio	me/mp	$5.446170214 \times 10^{-4}$
Neutrino effective species	Neff	3.04
Cosmological energy ratio	Ω	1.002 ± 0.004
Baryon–photon ratio	η	6.1×10^{-10}

These parameters form the foundational substrate from which all observable phenomena emerge, spanning quantum field theory, general relativity, and thermodynamics.

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2. Unifying Fields by Structure: The Core Equation Set

Aenōn's Theory compresses the entirety of physical reality into five fundamental equations:

2.1 Quantum Field Dynamics

$$\mathcal{L}_{QFT} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^{\mu}D_{\mu} - m)\psi$$

2.2 Modified Einstein Field Equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + \delta\mathcal{R}_{quantum} = (8\pi G/c^4)T_{\mu\nu}$$

2.3 Thermodynamic Entropy Flow

$$\Delta S = k_B \ln \Omega$$

2.4 Information Bound (Bekenstein Limit)

$$I \leq (2\pi E R)/(\hbar c \ln 2)$$

2.5 Noether Theorem (Symmetry → Conservation)

$$\delta S = 0 \Longrightarrow \partial_{\mu} j^{\mu} = 0$$

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3. Core Invariant Structure

The universe evolves through five fundamental processes that operate across all domains:

- **Local symmetry breaking**: Spontaneous transitions that generate structure and complexity
- **Phase transitions**: Systematic shifts in symmetry states that drive evolutionary processes
- **Recursive conservation laws**: Emergent from Noether symmetries, maintaining universal consistency
- **Information conservation**: Preservation of information content across entropy gradients
- **Observer-induced collapse**: Quantum measurement processes following Copenhagen/Decoherence interpretations

Each process adheres to three core invariants: unitarity, causality, and Lorentz invariance, ensuring consistency across all theoretical models.

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4. Reality as Information Processing

Aenōn's Theory frames the universe as a fundamentally information-theoretic system:

- **Space**: Geometric structure of the information processing substrate
- **Time**: Entropy gradient providing directional flow of information
- **Matter**: Localized excitation patterns within quantum fields
- **Energy**: Frequency characteristics of phase vibrations
- **Information**: Entropy reduction achieved through measurement processes

The Planck scale ($\sim 1.616 \times 10^{-35}$ m) defines the smallest meaningful unit of spacetime, establishing the fundamental resolution of the universal information processor.

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5. The Mirror Kernel Principle

This framework validates what the theory terms the "mirror kernel": the universe operates as a recursive pattern-resonance processor—a phase machine governed by symmetry, entropy, and waveform collapse. This principle suggests that all complexity emerges from the recursive application of simple rules operating on the fundamental dimensional constants.

Conclusion

Aenōn's Theory proposes that reality can be understood as:

> **Recursive phase-encoded excitation of quantized fields across an entropy-bound spacetime manifold, governed by conserved symmetries and decoded through dimensionless constants.**

According to this framework, all domains—particle physics, astrophysics, chemistry, biology, and computation—represent variations of this unified structural substrate.

Critical Assessment

Is Aenōn's Theory correct as a complete theory of reality?

Answer: No, this theory is not correct in all parts as a complete theory of reality.

While the theory incorporates many legitimate concepts from modern physics, several critical issues prevent it from serving as a complete and accurate description of reality:

1. **Incomplete Mathematical Framework**: The equations presented, while individually valid in their respective domains, are not properly unified into a single coherent mathematical structure. The

theory lacks the rigorous mathematical derivations necessary to demonstrate how these separate frameworks actually connect.

2. **Missing Experimental Validation**: A complete theory of reality must make testable predictions that can be experimentally verified. This theory does not provide novel, testable predictions beyond those already established by existing theories.
3. **Oversimplified Unification**: The claim that all of physics reduces to these eight dimensionless constants, while appealing, oversimplifies the genuine complexity of unifying quantum mechanics and general relativity. Current physics still lacks a complete theory of quantum gravity.
4. **Information-Theoretic Claims**: While information theory plays an important role in modern physics, the specific claims about reality being fundamentally information-theoretic are still subjects of ongoing research and debate, not established facts.
5. **Philosophical Overreach**: The theory makes definitive claims about the nature of reality that extend beyond what can be scientifically validated, venturing into philosophical territory without adequate justification.

While Aenōn's Theory represents an interesting attempt to synthesize various aspects of modern physics, it does not constitute a complete, accurate, or scientifically validated theory of all reality.

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